KAISER PERMANENTE®

Migraine and Tension Headache Guideline

Last guideline approval: January 2025

Guidelines are systematically developed statements to assist patients and providers in choosing appropriate health care for specific clinical conditions. While guidelines are useful aids to assist providers in determining appropriate practices for many patients with specific clinical problems or prevention issues, guidelines are not meant to replace the clinical judgment of the individual provider or establish a standard of care. The recommendations contained in the guidelines may not be appropriate for use in all circumstances. The inclusion of a recommendation in a guideline does not imply coverage. A decision to adopt any particular recommendation must be made by the provider in light of the circumstances presented by the individual patient.

This evidence-based guideline was developed by **the KP Medical Foundation.**

Major Changes as of January 2025

- When managing **medication overuse headache,** it is an option to continue taking acute medications until preventive medication has been titrated to a therapeutic dose. Previously, it was recommended that patients stop all acute headache medications.
- **Zinc** supplementation is no longer recommended for migraine prevention due to insufficient evidence and toxicity concerns.
- Preferred **CGRP medications** for both prevention and acute treatment have been updated based on formulary changes.

Medications That Are *Not* Recommended

- Advise **against** the use of **opioids and butalbital-containing medications** (e.g., Fiorinal, Floricet) for treatment of headaches.
- Advise **against** the use of **estrogen-containing oral contraceptive pills for migraine with aura** due to increased risk for stroke.
- There is insufficient evidence to support the use of **SSRIs, venlafaxine, gabapentin, zinc, or coenzyme Q10** for migraine prophylaxis.
- **Butterbur** is not recommended for migraine prophylaxis because it may contain a compound that is hepatotoxic and carcinogenic.

Background

This guideline includes diagnosis and treatment of the most common headache types that are managed in Primary Care:

- Tension headache
- Migraine headache, including menstrual migraine
- Medication overuse headache (also known as rebound headache)

Cluster headaches are excluded from this guideline because of their low prevalence in the general population and the severity of the symptoms. For patients with suspected cluster headaches, consider consulting with Neurology for evaluation and treatment.

Populations excluded from this guideline include pregnant individuals and children aged 13 years and younger.

Diagnosis

Red flag warning signs

For patients with a rapidly accelerating course, a recent history of head injury, or focal neurologic findings, consult with a neurologist or neurosurgeon.

Use the **SNOOP** mnemonic to identify red flag warning signs requiring immediate or urgent evaluation:

Systemic
- Conditions: malignancy, HIV, pregnancy
- Signs: fevers, sweats, rash, weight loss

Neurologic
- Symptoms: any neurologic symptoms other than classic aura (such as confusion or double vision)
- Signs: optic nerve edema, abnormal neurologic exam

Onset sudden (< 5 minutes)
Older than 50 years
Pattern change
- Change in type or quality of headache
- More than 50% increase in frequency or severity

Consider using the printable Patient Questionnaire for Headaches to evaluate patients for red flags.

When patients have no red flags or indications for imaging, ask them to gather more data on their headaches and schedule follow-up in Primary Care in 1 to 2 weeks to assess their response to empiric treatment. The SmartPhrase **.AVSHEADACHEDIARY** (at KPWA) or the SmartText **HEADACHE DIARY FOR ALL AGES PI CO** (at KPCO).and the handout Your Headache Log provide patients instructions for keeping a headache log and advice about when to seek immediate medical attention.

Differential diagnosis

Consider the following “can’t miss” headache causes at least once in your evaluation of a new-onset headache or a change in an existing headache pattern.

Create a concrete differential diagnosis to rule out “can’t miss” causes of headache using the **DATA C^2A^2N** save lives mnemonic:

Dissection (carotid or vertebral)
Arteritis (giant cell)
Thrombosis (dural venous)
Aneurysm (leak, expansion, or subarachnoid hemorrhage)
Carbon monoxide, **C**olloid cyst
Angle closure glaucoma, **A**ngina
Norepi neoplasm (pheochromocytoma)

Imaging

Order imaging only when your differential diagnosis supports it. Imaging should not be done solely for reassurance. High-end imaging (CT or MRI) for uncomplicated headache increases costs, radiation (CT), and anxiety for patients without improving quality of care. See “Headache” topics in the KPWA Radiology/ Imaging Quick Care Guide.

Most urgent causes of headache are **not** ruled out with a non-contrast head CT and need to be excluded with specific imaging, exam, or serologic testing. **A head CT does not “clear” a patient with headache.**

Migraine versus tension headache

Source: International Headache Society 2018

Table 1. Distinguishing between migraine and tension-type headache	
Migraine headache	**Tension headache**
4–72 hours' duration. Aura may be present. *At least two* of the following bullets are true: • Unilateral location. • Moderate to severe pain intensity. [1] • Pain described as pulsating. • Aggravated by routine activity. *At least one* of the following two bullets is true: • Sensitivity to light and/or sound is present. • Nausea and/or vomiting is present.	30 minutes' to 7 days' duration. No aura. *At least two* of the following bullets are true: • Bilateral location. • Mild to moderate pain intensity. [1] • Pain described as pressing or tightening (not pulsating). • Not aggravated by routine activity. *Both* of the following bullets are true: • No sensitivity to light or sound, **or** sensitivity to only one of the two. • Neither nausea nor vomiting is present.
1 Pain intensity • Mild: Patient is aware of headache, but able to continue daily routine with minimum alterations. • Moderate: Headache inhibits daily activities but is not incapacitating. • Severe: Headache is incapacitating.	

Medication overuse headache (MOH)

Source: International Headache Society 2018

Medication overuse headache (MOH; also known as *rebound headache)* is headache occurring at least 15 days per month in patients with pre-existing primary headache who have regularly overused acute or symptomatic headache medication for 3 months or longer. ("Overuse" is defined as > 10 days or > 15 days per month, depending on the medication**.** Acute or symptomatic headache medications include Tylenol, Ibuprofen or other NSAIDs, and Excedrin, as well as prescription headache medications, such as triptans.) It usually, but not invariably, resolves after the overuse is stopped. A common culprit is over-the-counter medications, which are not always on the medication list. *Note:* Use of acute drugs for non-headache purposes (e.g., ibuprofen for back pain) counts towards the total number of days used per month.

It is especially important to avoid the use of **opioids and butalbital-containing medications** (e.g., Fiorinal, Floricet) in the context of MOH; some population studies suggest patients may experience MOH after taking opioids and barbiturates < 10 days.

It is important to rule out MOH prior to initiating therapy for any acute headache. See MOH Treatment, p. 17.

Menstruation-related migraine

Source: International Headache Society 2018

Episodes of migraine without aura (as defined in Table 1) occurring in the window of 2 days before to 3 days after menstruation, in at least two out of three menstrual cycles.

Tension Headache

Acute treatment of tension headache

Note: It is important to rule out medication overuse headache (MOH) prior to initiating therapy for any acute headache. See MOH Treatment, p. 17.

Table 2. Pharmacologic options for acute treatment of tension headache				
NOT RECOMMENDED				
Do *not* use opioids and butalbital-containing medications (e.g., Fiorinal, Fioricet) for treatment of headaches.				
ASPIRIN/NSAIDS				
Medication	**Initial dose**	**Max dose/day**	**Relative *contraindications***	**Special considerations**
Aspirin	500 mg x1, may repeat in 4–6 hours	4000 mg	Age < 19 years Post–Roux-en-Y gastric bariatric surgery History of GI bleeding	OTC Possible side effects: GI
Ibuprofen	400 mg x1, may repeat in 4–6 hours	1200 mg	Post–Roux-en-Y gastric bariatric surgery History of GI bleeding	OTC Possible side effects: GI, cardiovascular, and renal
Acetaminophen/aspirin/ caffeine	500 mg (aspirin component), may repeat in 6 hours	4000 mg (acetaminophen component)	Age < 19 years Post–Roux-en-Y gastric bariatric surgery History of GI bleeding	OTC Ask about acetaminophen from other sources. Lower max dose in severe liver disease.
Naproxen	500 mg x1, may repeat in 6–8 hours	1250 mg	Post–Roux-en-Y gastric bariatric surgery History of GI bleeding	OTC Possible side effects: GI, cardiovascular, and renal
Acetaminophen	1000 mg, may repeat in 6 hours	4000 mg [1]	Liver disease	OTC
[1] For patients aged 65+ and patients with liver disease, maximum dose is 3000 mg.				

Prophylaxis of tension headache

Self-care

Advise the patient about the following self-care strategies:

- Keeping a regular schedule of sleep, exercise, and good nutrition. Poor sleeping and eating patterns are triggers for headaches.
- Rearranging work or study areas to avoid physical strain. For example, moving computer screens to eye level, lowering chair so that thighs are parallel to the floor, using a lumbar roll to maintain a good sitting posture, and using a phone headset instead of cradling phone on the neck.
- Gentle stretching exercises and relaxation techniques to prevent neck pain. Heat and ice to relieve neck pain and gentle stretches to help loosen tension in the neck. Robin McKenzie's book *Treat Your Own Neck* is a good source for effective self-care exercises to lower neck muscle tension naturally. Consider effects of depression and anxiety in neck tension.
- Kaiser Permanente offers free apps such as the Calm app to help patients incorporate self-care into their daily lives. Consider using **.DIGITALSELFCARE**. Despite the lack of strong evidence, mindfulness and good self-care are important ways to promote wellness.
- Limiting caffeine intake to no more than 2 cups a day to help avoid caffeine withdrawal headaches.
- Limiting use of over-the-counter pain medicines (such as acetaminophen, acetaminophen/aspirin/caffeine (Excedrin), aspirin, or ibuprofen) or decongestants (such as Sudafed or pseudoephedrine) to no more than 3 days a week for headaches, to avoid medication overuse headaches.

Monitoring and prophylaxis planning

- Advise the patient to keep and review a headache diary to monitor the effects of treatment on severity, frequency, and disability. Consider using **.AVSHEADACHEDIARY** (at KPWA) or **HEADACHE DIARY FOR ALL AGES PI CO** (at KPCO). Patients may opt to use a smartphone app, such as Migraine Buddy, an electronic headache diary.
- Work with the patient to establish an individualized goal of prophylaxis, noting that reducing the frequency and/or severity of headaches—rather than eliminating them completely—is a realistic target.
- Consider follow-up by phone visit in 4–6 weeks to check and adjust treatment options.

Complementary/alternative medicine

See Options for migraine prophylaxis: Complementary/alternative medicine (CAM), p. 11.

Medication

Amitriptyline is the preferred medication for prevention of tension headaches. If amitriptyline is not effective/ tolerated, a trial of any of the medications listed in Table 5 for prevention of migraines would be appropriate.

Migraine Headache

Acute treatment of migraine in Primary Care

The choice of acute migraine treatments should be dictated by the rapidity of onset, headache severity, associated symptoms (e.g., nausea/vomiting), and patient preference. If a patient doesn't respond to 1–2 adequate doses of a given medication during a migraine episode, it is appropriate to try another medication. Use **.MIGRAINEAVS** (at KPWA) or **.MIGRAINE PI CO** (at KPCO).

Note: It is important to rule out medication overuse headache (MOH) prior to initiating therapy for any acute headache. See MOH Treatment, p. 17.

Table 3. Pharmacologic options for acute treatment of migraine in Primary Care

NOT RECOMMENDED					
Do *not* use opioids and butalbital-containing medications (e.g., Fiorinal, Fioricet) for treatment of headaches.					
TRIPTANS • Triptans are first-line treatment for severe migraines as they are generally highly effective, with a low risk of side effects. • Failure of one triptan does not indicate failure of the entire class of medication. Consider trying a second triptan medication if the first one does not improve symptoms. • A combination of triptan and NSAID may be more effective than either medication alone.					
Medication	**Initial dose**	**Max dose/day**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Eletriptan - oral	20–40 mg x 1, may repeat in 2 hours	80 mg	—	Concomitant ergot or MAOI use Cerebrovascular syndrome Significant cardiovascular disease Hemiplegic or basilar migraine Severe hepatic impairment	Possibly the most effective triptan Contraindicated within 72 hours of moderate to strong CYP3A4 inhibitors
Sumatriptan - oral	25–100 mg x1, may repeat in 2 hours	200 mg	Relatively safe in pregnancy		Good choice for most people, most of the time Requires dose adjustment for hepatic impairment
Rizatriptan - oral	5–10 mg x1, may repeat in 2 hours	30 mg	—		Prescribe 5 mg dose with concomitant propranolol (max 15 mg/day)
Naratriptan - oral	1–2.5 mg x1, may repeat in 4 hours	5 mg	People with ≥ 3-day headaches (long half-life)		Complement to NSAIDs for perimenopausal patients Requires dose adjustment for renal and hepatic impairment
Zolmitriptan - oral	2.5 mg x 1, may repeat in 2 hours	10 mg	—		Requires dose adjustment for hepatic impairment
TRIPTANS: *non-preferred*					
Sumatriptan – nasal solution (e.g., Imitrex)	5–20 mg x1, may repeat in 2 hours	40 mg	Nausea/vomiting or rapid peak in migraine intensity	Concomitant ergot or MAOI use Cerebrovascular syndrome Significant cardiovascular disease Hemiplegic or basilar migraine	Taste not acceptable to some patients High cost
Sumatriptan – SQ	6 mg x1, may repeat in 1 hour	12 mg			High cost
Table 3 continues…					

Table 3 *continued.* Pharmacologic options for acute treatment of migraine in Primary Care					
ASPIRIN & NSAIDs					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Aspirin + acetaminophen + caffeine (Excedrin Migraine or similar)	500 mg (aspirin component), may repeat in 6 hours	4000 mg (acetaminophen component)	—	Age < 19 years Post–Roux-en-Y gastric bariatric surgery History of GI bleeding	OTC Ask about acetaminophen from other sources Lower max dose in severe liver disease
Naproxen	500 mg x1, may repeat in 6–8 hours	1250 mg	—	Post–Roux-en-Y gastric bariatric surgery History of GI bleeding	OTC Possible side effects: GI
Ibuprofen	400 mg x1, may repeat in 4–6 hours	1200 mg	—	Post–Roux-en-Y gastric bariatric surgery History of GI bleeding	OTC Possible side effects: GI
ANTIEMETICS					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Metoclopramide (Reglan)	5 mg x1, may repeat with adjunctive medication	20 mg	Nausea/vomiting	People at risk for extrapyramidal syndromes (EPS)	Adjunct only, not standalone Caution with long-term use
Prochlorperazine	5–10 mg every 6 to 8 hours as needed	40 mg			
ERGOTS					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Dihydroergotamine - nasal	0.5 mg x1, may repeat in 15 minutes	2 mg daily 4 mg weekly	More severe headache Nausea/vomiting	Safety/efficacy not established in pediatrics Pregnancy Hemiplegic or basilar migraine Ischemic heart disease Severe hepatic or renal impairment	After failure of preferred treatment High cost Possible side effects: ergotism (peripheral vascular ischemia, headache, vomiting, diarrhea, gangrene of the fingers and toes)
Dihydroergotamine – SQ/IM	1 mg x1, may repeat in 1 hour	3 mg daily 6 mg weekly			

Table 4. Additional pharmacologic options for acute treatment of migraine in Primary Care: ***Neurology consultation required***					
CGRP INHIBITORS Weak evidence supports the use of CGRP inhibitors for acute treatment of migraine. The team recommends keeping these medications as second-line options due to high cost.					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Ubrogepant (Ubrelvy)	50–100 mg x1, second dose may be taken ≥ 2 hours after initial dose	200 mg/24 hours	Contraindication(s) to triptan	Concomitant use with strong CYP3A4 inhibitors Concomitant use with strong CYP3A4 inducers Pregnancy End-stage kidney disease	High cost Frequent drug-drug interactions requiring dose adjustments Requires dose adjustment for renal and hepatic impairment Avoid using with CGRP monoclonal antibody See also CGRP inhibitor criteria for KPWA or KPCO.
CGRP INHIBITORS: ***non-preferred***					
Rimegepant (Nurtec)	75 mg x1	75 mg/24 hours	Contraindication(s) to triptan	Concomitant use with strong CYP3A4 inhibitors Concomitant use with moderate/strong CYP3A4 inducers Pregnancy End-stage kidney disease Severe liver impairment	High cost Frequent drug-drug interactions requiring dose adjustments Not advised to use as both acute and preventative therapy Avoid using with CGRP monoclonal antibody See also CGRP inhibitor criteria for KPWA or KPCO.

Treatment of refractory migraine

For patients with migraines that are not relieved by home care, consider treatment with **fluids, an antiemetic, and an NSAID.** Opioids are **not** recommended.

If a patient presents to ER or Urgent Care with a high-risk or unusual acute migraine care plan (e.g., one involving opioids or benzodiazepines), consider outpatient E-Consult with Neurology to determine a more evidence-based treatment plan.

The choice of medication should be directed by the severity of the attack, the type of symptoms present, patient preference, and patient-specific factors. Two possible starting points are:

- A “cocktail” of IV fluids, IV or IM ondansetron, and IM ketorolac
- A “cocktail” of medications that has worked for the patient before and does **not** include opioids

Sumatriptan SQ may be a useful adjunct if indicated. For other IV or IM options, consult with Urgent Care or Neurology.

For patients who have repeated refractory migraines, consider medication overuse as an underlying cause.

Migraine prophylaxis

Overview

There are no clear evidence-based recommendations for when to start preventive therapy for migraine.

The choice of migraine prevention medication should be made based on comorbid conditions (e.g., consider tricyclics for patients with depression and/or neuropathic pain syndromes, valproic acid/divalproex for persons with seizure disorders) and the relative value the patient places on efficacy versus avoidance of side effects.

Develop a **written headache treatment plan** for prevention and management of acute migraine to:

- Decrease headache frequency. (Aim for fewer than 5 headache days per month.)
- Decrease headache severity. (Headaches will respond quickly to an abortive therapy.)
- Avoid medication/caffeine overuse headache. (See MOH Treatment, p. 17.)

Guiding principles of migraine prophylaxis

- Each medication dose change may take 2–4 weeks to reach maximal effectiveness.
- Fewer than 5 headache days per month is the goal.
- The initial dose of each medication should be fairly low, and gradually increased to the maximal tolerated or maximal safe dose. Keep medication at that dose for 1 month before making modifications to therapy.
- Ideally, an adequate trial of a prophylactic medication is 8 weeks at the *maximum recommended or tolerated dose* before considering an alternative medication.
- Patients may need more than one prophylactic medication to decrease headache frequency. Set expectations with patients that it may take up to 3 months to find the most effective combination of medication and dosing.
- When headache days remain fewer than 5 per month for 1 to 2 months, start tapering therapy of the least well-tolerated medication to find the lowest effective dose.
- ***Note:*** It is important to address medication overuse headache (MOH) prior to initiating headache prophylaxis. See MOH Treatment, p. 17. Patients not responding to combinations of two or three prophylactic medications should be reassessed for MOH, an alternative diagnosis, or confounding psychiatric or social stresses. Drug therapy alone may not be sufficient.

Options for migraine prophylaxis

Self-care

Advise the patient about the following self-care strategies:

- Paying attention to food and beverages consumed, as they can trigger migraine attacks. Possible triggers include red wine (sulfites), aged cheeses, and chocolate.
- Keeping a regular schedule of sleep, exercise, and good nutrition. Poor sleeping and eating patterns are triggers for headaches.
- Rearranging work or study areas to avoid physical strain. For example, moving computer screens to eye level, lowering chair so that thighs are parallel to the floor, using a lumbar roll to maintain a good sitting posture, and using a phone headset instead of cradling phone on the neck.
- Gentle stretching exercises and relaxation techniques to prevent neck pain. Heat and ice to relieve neck pain and gentle stretches to help loosen tension in the neck. Robin McKenzie's book *Treat Your Own Neck* is a good source for effective self-care exercises to lower neck muscle tension naturally. Consider effects of depression and anxiety in neck tension.
- Kaiser Permanente offers free apps such as the Calm app to help patients incorporate self-care into their daily lives. Consider using **.DIGITALSELFCARE** (at KPWA and KPCO). Despite the lack of strong evidence, mindfulness and good self-care are important ways to promote wellness.
- Limiting caffeine intake to no more than 2 cups a day to help avoid caffeine withdrawal headaches.
- Limiting use of over-the-counter pain medicines (such as acetaminophen, acetaminophen/aspirin/caffeine (Excedrin), aspirin, or ibuprofen) or decongestants (such as

Sudafed or pseudoephedrine) to no more than 3 days a week for headaches, to avoid medication overuse headaches.

Monitoring and prophylaxis planning

- Advise the patient to keep and review a headache diary to monitor the effects of treatment on severity, frequency, and disability. Consider using **.AVSHEADACHEDIARY** (at KPWA) or **HEADACHE DIARY FOR ALL AGES PI CO** (at KPCO). Patients may opt to use a smartphone app such as Migraine Buddy, an electronic headache diary.
- Work with the patient to establish an individualized goal of prophylaxis, noting that reducing the frequency and/or severity of headaches—rather than eliminating them completely—is a realistic target.
- Consider follow-up by phone visit in 4–6 weeks to check and adjust treatment options.

Complementary/alternative medicine (CAM)

Note: Patients may incur out of pocket costs for these CAM treatments. For questions about coverage, patients can contact Member Services. Lists of preferred CAM providers:

- **KPWA:** KPWA member website at https://wa-member.kaiserpermanente.org/html/public/services/alternative (log-in required).
- **KPCO:** Kaiser Permanente Centers for Complementary Medicine website at http://www.kpccm.org/ or use **NEURO MIGRAINES COMPLMENTARY MED CLINRX PI CO** SmartText.

Acupuncture

- Moderate evidence supports the effectiveness of acupuncture in reducing migraine frequency and medication use in adult patients who have not responded to prophylactic medications.
- There is no evidence on the impact of acupuncture on migraine severity.
- There is no evidence on the effectiveness of acupuncture in teens.

Biofeedback

- Biofeedback is a relaxation technique that uses special equipment to teach the patient how to monitor and control certain physical responses. The most common types of biofeedback are thermal (hand-warming) and electromyographic (EMG).
- Low-strength evidence suggest that biofeedback may help decrease the frequency, duration, and intensity of both migraine and tension-type headaches.
- Biofeedback may also improve secondary outcomes of headache, such as medication use, muscle tension, anxiety, and depression.

Cognitive behavioral therapy (CBT)

- CBT teaches patients strategies on how to cope with everyday stressors that may bring on, worsen, or prolong headaches.
- Low-strength evidence suggests that including a CBT component in self-management interventions may double the effect size on mood but has no significant effect on headache-related disability or pain intensity, so will be most helpful for patients who are also experiencing depression or anxiety.

Mindfulness

- Mindfulness is a mental state achieved by focusing one's awareness on the present moment, while calmly acknowledging and accepting one's feelings, thoughts, and bodily sensations.
- The evidence on use of mindfulness as a therapeutic intervention for migraine or tension-type headache is conflicting.
- Due to lack of harms, mindfulness may be considered for patients who are interested in alternative treatments.
- Mindfulness may be most helpful for patients who are also experiencing depression or anxiety.

Yoga

- Low-strength evidence suggests that yoga may reduce headache frequency, duration, and pain intensity in patients with chronic or episodic tension-type headache.

- When used as an adjunct to medication treatment, yoga + medications may be more effective than medication alone.

Relaxation training

- Relaxation techniques include deep breathing exercises, progressive muscle relaxation, mental imagery relaxation, and relaxation to music.
- There is insufficient published evidence to recommend relaxation training for the treatment of adults with migraine or tension headache.

Neuromodulation devices

Note: While there is insufficient evidence to recommend for or against any form of neuromodulation for the treatment and/or prevention of migraine, these devices may be an option for patients who would like to avoid taking medication. Neuromodulation devices—such as Cefaly, GammaCore, and Nerivio— are not covered by insurance, but patients may use their Health Savings Account to purchase if desired.

Cefaly device

- One high-quality randomized controlled trial indicates that transcutaneous supraorbital stimulation using the Cefaly device may be more effective than sham procedure in the short term as prevention therapy.
- The long-term effect of using a Cefaly device is not known.
- Prescription is not required. Patients may order through the product website.

Supplements

For more information on over-the-counter supplements, see Consumer Reports or ConsumerLab.com.

Riboflavin (B2)

- Moderate-strength evidence supports the effectiveness of riboflavin in reducing migraine frequency, duration, and disability in the short term (3 months).
- The recommended dose of riboflavin is 400 mg daily.
- The most common side effects are polyuria and diarrhea.
- Consider in combination with magnesium.
- The onset of therapeutic action is 1 to 3 months.

Magnesium

- Low-strength evidence supports the use of magnesium in migraine prophylaxis.
- The recommended dose is 600 mg of elemental magnesium daily.
- Possible side effects include stomach upset and diarrhea.

Vitamin D

- Vitamin D may be beneficial in reducing the frequency of migraine headache days in patients with migraines **and** a known history of vitamin D deficiency. However, vitamin D screening is not routinely recommended.
- Vitamin D does not reduce the severity, duration, or associated symptoms of migraines.
- The dose of vitamin D used in a single randomized controlled trial was 4000 IU/day (Gazerani 2019). No clinical trials have examined the influence of vitamin D supplementation on tension type headaches.

Melatonin

- Melatonin may be more effective than placebo in preventing migraines.
- The recommended dose of melatonin is 3 mg.
- Is it difficult to separate the confounding role of melatonin in mood disorders and sleep problems from its impact on migraines.
- There is insufficient evidence to determine the comparative effectiveness of melatonin versus other therapies for preventing migraines in adults.

Medications and procedures

Table 5 – Medication options for migraine prophylaxis (pp. 12–13)
Table 6 – Additional procedure and medication options for migraine prophylaxis: *Neurology consultation required* (p. 14)

Table 5. Medication options for migraine prophylaxis in Primary Care

NOT RECOMMENDED

Do *not* use opioids and butalbital-containing medications (e.g., Fiorinal, Fioricet) for treatment of headaches.
There is **insufficient evidence to support** the use of SSRIs, venlafaxine, gabapentin, or coenzyme Q10 as prophylactic treatment for migraine.
Butterbur is not recommended because it may contain a compound that is hepatotoxic and carcinogenic.

BETA BLOCKERS

Medication	Initial dose	Max dose	Relative *indications*	Relative *contraindications*	Special considerations
Propranolol IR Propranolol ER	40 mg b.i.d. (IR) 80 mg daily (ER)	240 mg (divided t.i.d.) (IR) 240 mg daily (ER)	Hypertension Angina Essential tremor Stress-related migraine Tachycardia	Asthma/COPD Depression CHF Raynaud's disease Diabetes Bradycardia	Dose should be titrated and maintained for at least 3 months before assessment of response. See "Guiding principles of migraine prophylaxis" above.
Metoprolol IR Metoprolol ER	25 mg b.i.d. (IR) 25-50 mg daily (ER)	200 mg (divided b.i.d.) (IR) 200 mg daily (ER)	Hypertension Angina Atrial fibrillation CHF Myocardial infarction Tachycardia	Depression Raynaud's disease Diabetes Bradycardia	Titrate slowly (e.g., every 1 to 2 weeks)

ACE/ARB INHIBITORS

Medication	Initial dose	Max dose	Relative *indications*	Relative *contraindications*	Special considerations
Lisinopril	5 mg	40 mg	Intolerance to other migraine meds Hypertension ASCVD CHF	Pregnancy Hereditary or idiopathic angioedema	Cough is a common side effect Requires dose adjustments for renal and hepatic impairment
Candesartan	4–8 mg once daily	16 mg once daily	Hypertension ASCVD CHF		Non-formulary at KPWA Requires dose adjustments for renal and hepatic impairment

Table 5 continues…

Table 5, *continued*. Medication options for migraine prophylaxis in Primary Care

Medication	Initial dose	Max dose	Relative *indications*	Relative *contraindications*	Special considerations
ANTIEPILEPTICS					
Topiramate	25 mg daily	100 mg (divided b.i.d.)	Seizure disorder Bipolar disorder Obesity	Pregnancy Kidney stones Weight loss concerns Cognitive impairment Glaucoma	Titrate by 25 mg weekly Monitor serum bicarbonate, BUN, Cr while titrating Consider in obese patients Possible side effects: • "Brain fog" • Weight loss • Metabolic acidosis
Divalproex DR	125 mg b.i.d.	2000 mg (divided b.i.d.)	Seizure disorder Bipolar disorder Mania	Pregnancy Liver disease Bleeding disorder Obesity	Titrate by 125 mg weekly Monitor ALT, AST, CBC while titrating Need for highly effective contraception (e.g., LARC or two methods) Possible side effects: • Weight gain or loss • Hair loss • Tremor • Pancreatitis
TRICYCLIC ANTIDEPRESSANTS					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Nortriptyline Amitriptyline	10–25 mg q.h.s.	150 mg	Neuropathic pain Depression Anxiety Insomnia	Mania Urinary retention Heart block High risk in aged 65+	*Nortriptyline* Take at bedtime May be helpful for neck tightness, occipital myalgia, insomnia Better tolerated than amitriptyline *Amitriptyline* Effective combined w/topiramate

Table 6. Additional options for migraine prophylaxis in Primary Care: *Neurology consultation required*					
MEDICATIONS (anti-CGRP monoclonal antibodies) Strong evidence supports the use of CGRP inhibitors as safe and effective for the **prevention** of migraine. The team recommends keeping these medications as second-line options due to high cost.					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Fremanezumab-vfrm (SQ) (Ajovy)	225 mg monthly **or** 675 mg every 3 months	225 mg monthly **or** 675 mg every 3 months	—	Pregnancy	Long-term safety unknown High cost Recommend trialing Ajovy with monthly injections first Ajovy 675 mg dose is completed by injecting 3 syringes consecutively on the same day Adequate trial is 3 months on monthly injections and 6 months on quarterly injections **KPWA:** Does not require formal referral. Can be prescribed by Primary Care after E-Consult with Neurology. See KPWA criteria **KPCO:** Must be prescribed by a neurologist. See KPCO criteria.
PROCEDURES					
Procedure	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Occipital nerve block	NA	NA	May reduce the frequency and intensity of migraine headache Can be used as an adjunct treatment	—	Administered by Neurology
Botulinum toxin (onabotulinumtoxinA)	NA		Limited benefit, reduces chronic migraine frequency by around 2 days/month and improves patient quality of life when compared to placebo	—	Administered by Neurology Should be given every 3 months **KPWA:** Adequate trial of at least 3 formulary-preferred prophylactic migraine medications may be required for health plan coverage. **KPCO:** No PA criteria; however, must be administered by Neurology provider.

Menstruation-related migraine prophylaxis

Note: For acute treatment of menstruation-related migraine, see "Acute treatment of migraine in Primary Care" on p. 7.

Monitoring and prophylaxis planning

- Advise an individual with suspected menstrual migraine to keep a headache diary for at least 2 months to determine if they are having regular and predictable menses and if these correlate with headache onset. Some patients experience migraines around the time of ovulation (either alone or in addition to during menses).
- Use **.AVSHEADACHEDIARY** (at KPWA) or **HEADACHE DIARY FOR ALL AGES PI CO** (at KPCO). Patients may opt to use a smartphone app, such as Migraine Buddy, an electronic headache diary.
- Work with the patient to establish an individualized goal of prophylaxis, noting that reducing the frequency and/or severity of headaches—rather than eliminating them completely—is a realistic target.
- Consider follow-up by phone visit in 4–6 weeks to check and adjust treatment options.

Medication options

Table 7. Medication options for menstruation-related migraine prophylaxis

NOT RECOMMENDED					
Do *not* use estrogen-containing **oral contraceptive pills** for menstrual migraine **with aura** due to increased risk for stroke. **Do *not* use opioids and butalbital-containing medications** (e.g., Fiorinal, Fioricet) for treatment of headaches.					
ORAL CONTRACEPTIVES Oral contraceptives are first-line treatment for patients with irregular menses and premenstrual migraines **without aura.** Aura is defined as non-headache symptoms (e.g., tingling in limbs, visual disturbances) that precede the onset of a migraine headache.					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Use shared decision-making to choose appropriate option for the individual. Consider using Oral Contraceptive SmartRx (at KPWA) or OCP SMARTRX CO (at KPCO).			—	Migraine with aura	—
NSAIDS Consider a short course of prophylaxis with naproxen or a triptan twice a day starting 2–3 days before the anticipated onset of the headache and continuing for 5 days through the at-risk period.					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Naproxen	500 mg x1, may repeat in 6–8 hours	1250 mg	—	Post–Roux-en-Y gastric bariatric surgery	OTC Possible GI side effects
TRIPTANS Long-acting forms are recommended. Consider a short course of prophylaxis with naproxen or a triptan twice a day starting 2–3 days before the anticipated onset of the headache and continuing for 5 days through the at-risk period.					
Medication	**Initial dose**	**Max dose**	**Relative *indications***	**Relative *contraindications***	**Special considerations**
Naratriptan	1–2.5 mg x1, may repeat in 4 hours	5 mg	Patients with ≥ 3-day headaches (long half-life)	Concomitant ergot or MAOI use Cerebrovascular syndrome Significant cardiovascular disease Hemiplegic or basilar migraine	Complement to NSAIDs for perimenopausal individuals Prescribe 1 mg dose in mild to moderate renal or hepatic disease
Zolmitriptan	1.25–2.5 mg x1, may repeat in 2 hours	5 mg	Patients with ≥ 3-day headaches (long half-life)		—

Medication Overuse Headache Treatment

Medication overuse headache (MOH) is a state of daily or near-daily refractory headaches. Also known as *rebound headache,* MOH occurs at least 15 days per month in patients with pre-existing headache disorder who have regularly overused acute or symptomatic headache medication for 3 months or longer. (“Overuse” is defined as > 10 days or > 15 days per month, depending on the medication.)

Rule out medication overuse headache prior to initiating therapy for any acute headache. If MOH is present, prevention and acute headache medications may not be effective. Overuse is a concern with all headache medications, including over-the-counter medications (all NSAIDS, analgesics, or decongestants) and caffeine in more than moderate amounts. Overuse of prescription acute migraine remedies (e.g., sumatriptan, rizatriptan) can also cause medication overuse headaches.

For patients found to have MOH,

1. Advise that it will take 2–6 weeks to resolve the MOH after withdrawing all acute headache medications that the patient has been taking.
2. Start a prophylactic medication, ideally prior to stopping acute headache medications. It is an option to continue taking acute medications until the preventive medication is effective; patients will naturally taper use of acute medications as frequency of headaches decreases. The long-term goal is for patients to taper the acute medications down to 2 days/week or less. It is also an option to completely discontinue acute medications when starting preventive medication.
3. See “Options for migraine prophylaxis—medications and procedures,” p. 13.
4. Treat symptoms during withdrawal of acute headache medications.
5. Use **.AVSHEADACHEMEDOVERUSE** (at KPWA) or **.HEADACHE MED OVERUSE PI** CO (at KPCO) to develop a written headache treatment plan that includes acute and prophylactic treatment as well as a plan for close follow-up and relapse prevention.

Evidence Summary

The Migraine and Tension Headache Guideline was developed using an evidence-based process, including systematic literature search, critical appraisal, and evidence synthesis.

As part of our improvement process, the KP Medical Foundation guideline team is working towards developing new clinical guidelines and updating the current guidelines every 2–3 years. To achieve this goal, we are adapting evidence-based recommendations from high-quality national and international external guidelines, if available and appropriate. The external guidelines should meet several quality standards to be considered for adaptation. They must: be developed by a multidisciplinary team with no or minimal conflicts of interest; be evidence-based; address a population that is reasonably similar to our population; and be transparent about the frequency of updates and the date the current version was completed.

In addition to identifying the recently published guidelines that meet the above standards, a literature search was conducted to identify studies relevant to the key questions that are not addressed by the external guidelines.

External guidelines eligible for adapting

- 2023 VA/DoD Clinical Practice Guideline for the Management of Headache
- 2024 CGRP Position Statement of American Headache Association
- 2021 American Headache Society Consensus Statement: Update on integrating new migraine treatments into clinical practice.
- 2017 Hormonal contraceptives and risk of ischemic stroke in women with migraine: a consensus statement from the European Health Federation and the European Society of Contraception and Reproductive Health
- 2018 International Headache Society International Classification of Headache Disorders, 3rd Edition

Key questions

1. **What is the effectiveness, safety, and comparative effectiveness of CGRP inhibitors in the prevention of migraine?**
   New studies indicate that CGRP inhibitors are effective and safe for the prevention of migraine. The recommendations of the 2023 VA DOD guidelines should be adapted.
2. **What is the effectiveness, safety, and comparative effectiveness of CGRP inhibitors in the acute treatment of migraine?**
   New studies indicate that CGRP inhibitors are effective and safe for the acute treatment of migraine. The recommendations of the 2023 VA DOD guidelines should be adapted.
3. **What is the effectiveness of yoga, biofeedback, acupuncture, vitamin D, zinc, and melatonin in the prevention of migraine or tension-type headache?**
   For melatonin, zinc, and vitamin D, no high-quality studies challenge the current KPWA guidelines. For biofeedback, acupuncture, and yoga, no high-quality studies challenge the 2023 VA recommendations.
4. **What is the safety and effectiveness of home-based devices (Cephaly, GammaCore, eNeura, Nerivio) for treatment and/or prevention of migraine or tension-type headache?**
   The 2023 VA DOD guideline is adopted, as no new studies challenge its recommendations.
5. **What is the safety and effectiveness of serotonin 5-HT 1F (Lasmiditan) receptor agonists for the acute treatment of migraine headache?**
   No high-quality studies were identified. The literature (from 2022 to July 2024) is full of low-quality, post hoc analysis or extensions of randomized controlled trials. The 2023 VA DoD is adopted.

6. **What are the risks/harms associated with the use of estrogens in patients with migraine with aura?**
   There are few studies (from 2017 to 2024) assessing the risk of stroke in women with migraine who use estrogen. Available studies are poor and of low quality. Low-quality evidence from a systematic review (Sheikh 2018) and a case control study (Batur 2023) suggest an increased risk of stroke in women with migraine who use combined hormonal contraception. However, the results from the case control were surprisingly limited to those with migraine without aura. The consensus statement from the European Headache Federation (EHF) and the European Society of Contraception and Reproductive Health (ESC) (Sacco 2017) is adopted.
7. **Is medication withdrawal effective in the management of medication overuse headache?**
   The 2023 VA DOD guideline is adopted: "There is insufficient evidence to recommend for or against withdrawal strategy to guide the treatment of medication overuse headache."

References

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Guideline Development Process and Team

Development process

To develop the Migraine and Tension Headache Guideline, the guideline team adapted recommendations from external developed evidence-based guidelines and/or recommendations organizations that establish community standards. The guideline team reviewed additional evidence in several areas. For details, see Evidence Summary and References.

This edition of the guideline was approved for publication by the Guideline Oversight Group in December 2024.

Team

The Headache Guideline development team included representatives from the following specialties: Family Practice, General Internal Medicine, Neurology, Pharmacy, Residency, and Urgent Care.

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